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On Causal Representation Learning with Internal Auxiliaries

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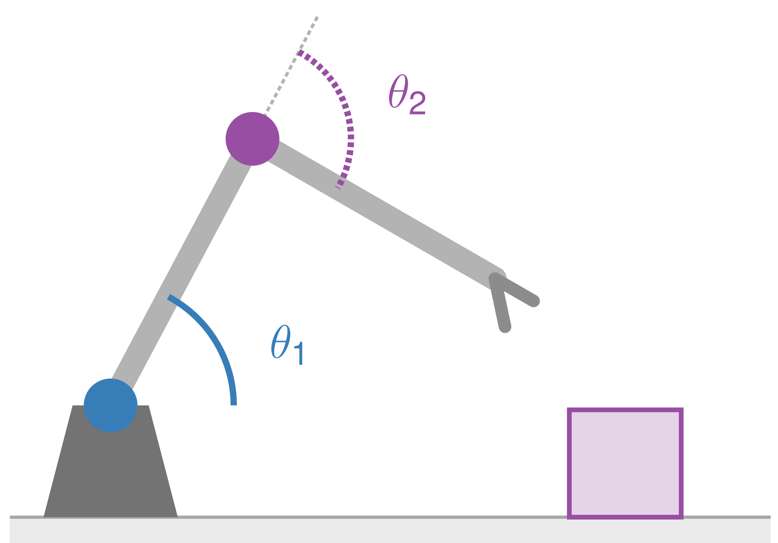
The auxiliary is *inside* the picture



- θ_1 **shoulder angle** — encoder ⇒ **OBSERVED**
- θ_2 **elbow angle, object pose** — *unobserved*

$$\mathbf{x} = g(\underbrace{\theta_1}_{\text{known}},\ \underbrace{\theta_2,\ \text{object pose}}_{\text{to be recovered}})$$

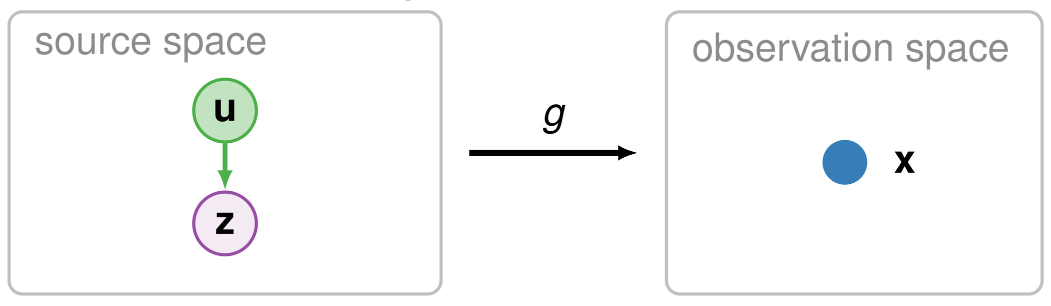
θ_1 is *measured* — but it is also **one of the coordinates the renderer consumed**, not a label sitting *outside* the picture. Standard auxiliary-based CRL assumes exactly that outside position. **An internal auxiliary breaks its proof.**

Internal auxiliary. An observed *pre-mixing source coordinate* $\mathbf{o} = \mathbf{s}_O$ — available at inference *and* an input to the mixing map: $\mathbf{x} = g(\mathbf{o}, \mathbf{z})$, with targets $\mathbf{z} = \mathbf{s}_{O^-}$.

Why the standard proof breaks

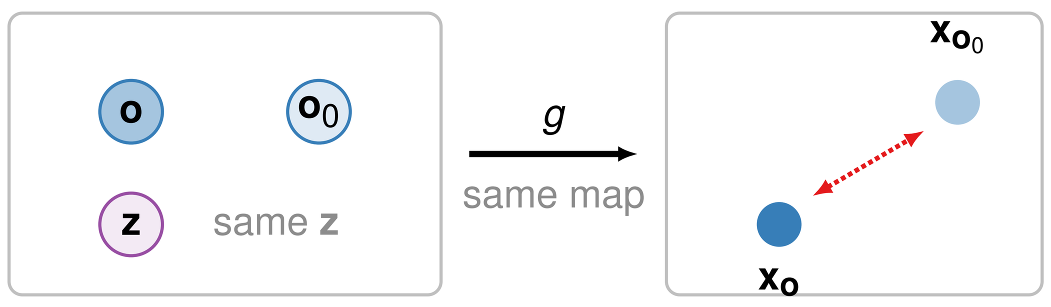
Let $\ell_g(\mathbf{x}) \equiv \log|\det \mathbf{J}_{g^{-1}}(\mathbf{x})|$ be the nuisance log-volume term. **One globally fixed** g **in both panels** — only *where we sit* changes.

(a) External auxiliary u



u has **no edge into x**: changing **u** leaves $\mathbf{x} = g(\mathbf{z})$ put. Both contexts are read at the **same x**, so ℓ_g **cancels on subtraction**.

(b) Internal auxiliary o (ours)



o is an input to g: changing **o moves the observation itself**. The contexts land on **different x** — nothing cancels.

The obstruction. Subtracting the two contexts leaves behind

$$\Delta_{\mathbf{o}}(\ell_{\hat{g}} - \ell_g) = \big[\ell_{\hat{g}}(\mathbf{x}_{\mathbf{o}}) - \ell_g(\mathbf{x}_{\mathbf{o}})\big] - \big[\ell_{\hat{g}}(\mathbf{x}_{\mathbf{o}_0}) - \ell_g(\mathbf{x}_{\mathbf{o}_0})\big] \neq 0.$$

⇓ **our fix**

Volume-preserving (VP) mixing. Restrict *both* maps: $|\det \mathbf{J}_g| = |\det \mathbf{J}_{\hat{g}}| = 1$, so $\ell_g \equiv \ell_{\hat{g}} \equiv 0$.

Each term is now zero **on its own** — we never subtract two contexts at all. VP does not *restore* the cancellation; it makes it **unnecessary**.

VP is a **sufficient assumption on the model class**, not a property of renderers.

Identifiability (Prop. 1)

THEOREM: $O = C$

$\{\mathbf{z}_{c_j}\}_{j=1}^{d}$ partitions the targets; $m_j = \dim(\mathbf{z}_{c_j})$, $q = \dim(\mathbf{z})$, $M = q + \sum_j m_j(m_j+1)/2$.

A1 Every observed internal source is conditioned on ($O = C$, so $\mathbf{o}_n = \emptyset$); the learned model is *observed-coordinate preserving*, $\hat{g}^{-1}(g(\mathbf{o}_C, \mathbf{z})) = (\mathbf{o}_C, \hat{\mathbf{z}})$; and *both* sides factorize, $p(\mathbf{z} \mid \mathbf{o}_C) = \prod_j p(\mathbf{z}_{c_j} \mid \mathbf{o}_C)$.

A2 VP mixing on both g and $\hat{g}$.

A3 Sufficient variability: M values of $\mathbf{o}_C$ give linearly independent score-curvature vectors (block gradients *and* vech of the block Hessians).

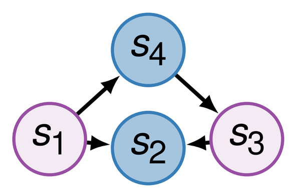
A4 Regularity (App.): common support, smooth maps, a *conditioning-independent* reparameterization $\mathbf{z} = r(\hat{\mathbf{z}})$, block matching, and Markov + faithfulness.

Then the unobserved subspaces are identifiable up to **permutation across subspaces** + **an invertible transform within each** (ISA-style class).

The theorem is a **square map** ($m = n$). The image results add an encoder/compression stage, so read them as **empirical**, not as instances of Prop. 1.

Choosing what to condition on

Conditioning on *more* observed sources can make the partition **coarser**: a conditioned collider **opens** a path.



blue = observed, $O = \{s_2, s_4\}$; purple = targets, $\mathbf{z} = \{s_1, s_3\}$. s_2 is a **collider**: $s_1 \to s_2 \leftarrow s_3$.

Condition on $\mathbf{o}_C$	Groups	
$\{s_2\}$	$\{s_1, s_3\}$	collider opened
$\{s_2, s_4\}$	$\{s_1, s_3\}$	"use everything" = worst
$\{s_4\}$	$\{s_1\}, \{s_3\}$	**finest** ✓

Alg. 1. Enumerate *all* nonempty $T \subseteq O$; score each by how many conditionally independent target groups it induces (d-separation); return the **smallest** set with the **largest** count.

Exhaustive $2^{|O|} - 1$, **no pruning** — the same node can be a collider on one path and a confounder on another. We demonstrate the **criterion**, not a scalable search.

NOT A THEOREM: $\mathbf{o}_n \neq \emptyset$

The winner $\mathbf{o}_C = \{s_4\}$ leaves $\mathbf{o}_n = \{s_2\}$ **observed but unselected** — **outside Prop. 1**, which assumes $O = C$. We handle $\mathbf{o}_n$ with the graph-structured term in the objective (→ **Model**); extending the theorem is **open**.

Model

Encoder: GIN — a volume-preserving flow, which is what enforces **A2**.

$$\begin{aligned}\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}}\Big[\ &\log p_{\text{align}}(\hat{\mathbf{o}}_C \mid \mathbf{o}_C) \\ &+ \textstyle\sum_j \log p(\hat{\mathbf{z}}_{c_j} \mid \mathbf{o}_C) \\ &+ \textstyle\sum_{i \in O \setminus C} \log p(o_i \mid \hat{f}_i(\hat{\mathbf{s}}_i^{\text{slot}}))\ \Big]\end{aligned}$$

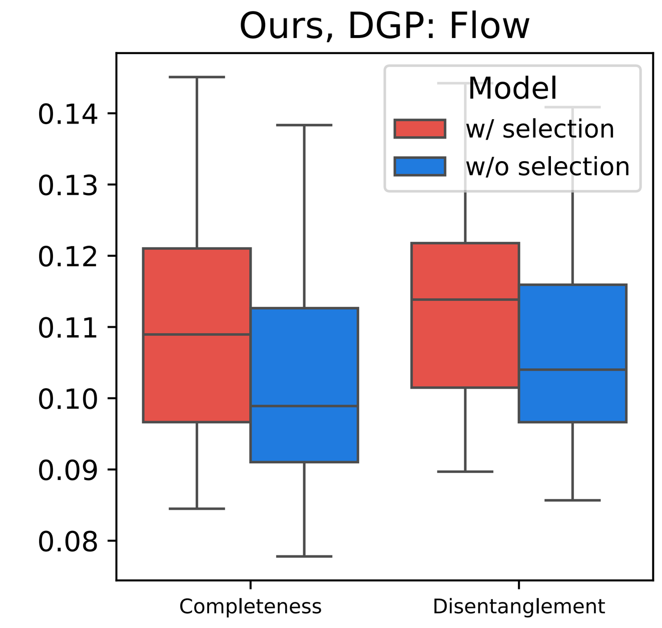
Align $\hat{\mathbf{o}}_C$ to $\mathbf{o}_C$; make the target subspaces conditionally independent given $\mathbf{o}_C$; predict each *unselected* o_i from its **graph-parent slots**. **Line 3, not the theorem, is what covers** $\mathbf{o}_n \neq \emptyset$.

Evidence

Baselines **GIN** and **iVAE** treat internal sources as *external* conditioners. This is a **mismatch test** by construction — not a SOTA sweep.

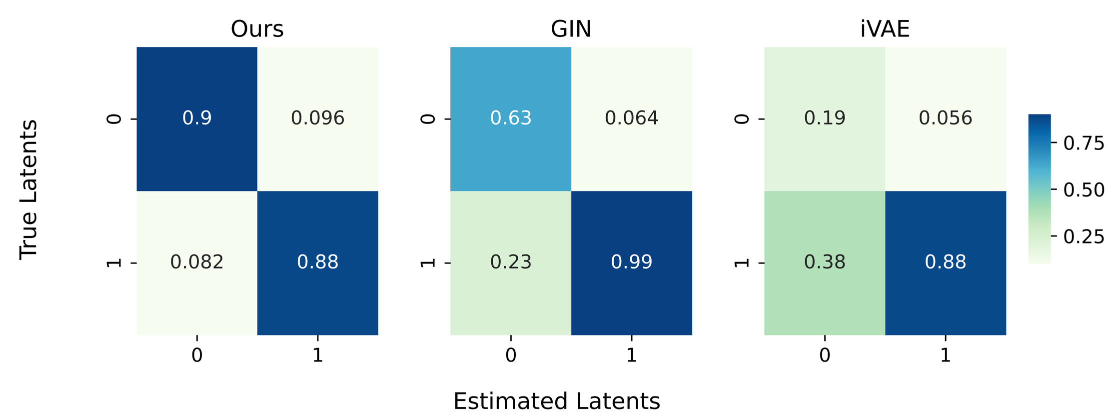
1. The selection criterion matters.

The only experiment that tests the graph claim directly.



Flow DGP, with vs. without selection. **Median improves on both metrics** (distributions overlap). Conditioning on *all* observed sources opens the **Water Height–Hole** collider path.

2. Less cross-factor leakage.



Pendulum, 4 × 96 × 96 images; MCC matrices, best-permutation matched. Axes: 0 = Shadow Length, 1 = Shadow Position.

Worst off-diagonal — **ours** .096 vs. .23 (GIN), .38 (iVAE). Evidence for **less leakage**; *not* a claim of identifiability on pixels.

3. Across all three synthetic DGPs, our DCI-style scores have the **higher median in all six** metric × DGP panels (20 seeds; distributions overlap). See the paper.

Takeaway

Internal auxiliaries **break** the context-comparison proof.

Under VP mixing, a **selected** internal source identifies the conditional factorization it induces — and a **known graph** chooses conditioning sources that **expose**, rather than destroy, the finest such factorization.

The graph is **assumed known**; we do not do causal discovery.

snu-causality-lab/crl-internal-auxiliaries
arXiv:2509.19058